

Department of Mathematics and Applied Mathematics
Module 3MS: Metric Spaces

Class test 2

02 May 2024

Time : 18:00 – 20:00

Total marks: 40

1. State whether each of these statements is true or false. No proof required.
 - (a) The function $f : (\mathbb{R}, d_D) \rightarrow (\mathbb{R}, d_E)$ defined by $f(x) = -x$ is an isometry.
 - (b) The function $f : (\mathbb{R}, d_D) \rightarrow (\mathbb{R}, d_E)$ defined by $f(x) = -x$ is continuous.
 - (c) The function $g : (\mathbb{N}, d_D) \rightarrow (\mathbb{N}, d_E)$ defined by $g(x) = x$ is a metric isomorphism.
 - (d) The function $g : (\mathbb{N}, d_D) \rightarrow (\mathbb{N}, d_E)$ defined by $g(x) = x$ is a topological isomorphism.
 - (e) The interval $[1, 95]$ is a complete subset of $(\mathbb{R}, d_E)$.
 - (f) The interval $[1, 95]$ is a complete subset of $(\mathbb{R}, d_D)$.
 - (g) The empty set $\emptyset$ is a compact subset of (any) metric space (X, d) . [7]

2.
 - (a) Prove: in any metric space, every convergent sequence is Cauchy.
 - (b) Prove or disprove: every closed subset of a complete metric space is complete.
 - (c) Prove or disprove: every closed subset of a complete metric space is compact. [7]

3. Let X be the set of positive integers. You are given, and do not need to check, that ρ is a metric on X , where
$$\rho(k, \ell) = \left| \frac{1}{k^2} - \frac{1}{\ell^2} \right|.$$
 - (a) Let $x_n = n$. Is (x_n) a Cauchy sequence in (X, ρ) ? Prove your claim.
 - (b) Is the metric space (X, ρ) complete? Prove your claim. [6]

P.T.O

4. Read the following theorem and proof, and then answer the questions that follow.

Theorem Let (X, d) be a complete metric space and $f : (X, d) \rightarrow (X, d)$ be a contraction mapping. Then f has a fixed point.

Proof Suppose that f is a contraction with Lipschitz constant k . (1)

Define (x_n) by taking an arbitrary $x_1 \in X$ and letting $x_{n+1} = f(x_n)$ for $n \geq 1$. (2)

It can be shown that $d(x_{n+1}, x_n) \leq k^{n-1}d(x_2, x_1)$. (3)

Then

$$m > n \implies d(x_n, x_m) \leq \frac{k^{n-1}}{1-k}d(x_2, x_1). \quad (4)$$

So (x_n) is a Cauchy sequence. (5)

Thus $x_n \rightarrow x$, for some $x \in X$. (6)

So $f(x_n) \rightarrow f(x)$. (7)

But $x_{n+1} \rightarrow x$. (8)

So $f(x) = x$, since limits are unique. (9)

- (a) What condition must k satisfy to ensure that f is a contraction?
- (b) Show how line (3) follows from the previous working.
- (c) Show how the claim that (x_n) is Cauchy follows from line (4).
- (d) How do we know that (x_n) converges, as claimed in line (6)?
- (e) How do we know that $f(x_n) \rightarrow f(x)$, as claimed in line (7)?
- (f) Explain what is meant by the expression “limits are unique” (in line (9)) and prove it. [11]

5. Suppose that d and ρ are metrics on the same set X , such that, for all $x, y \in X$,

$$d(x, y) \leq \rho(x, y) \leq 4d(x, y)$$

- (a) Prove that, for any sequence (x_n) in X and $x \in X$, $x_n \rightarrow x$ with respect to d if and only if $x_n \rightarrow x$ with respect to ρ .
- (b) State two facts about the spaces (X, d) and (X, ρ) , which you are able to deduce from (a) above. Just write down the statements; no proofs required. [7]

6. We define a metric space (X, d) to have “Property P” if it satisfies the following condition:

For every $\epsilon > 0$, there exists a finite subset F of X such that $\bigcup_{t \in F} B(t; \epsilon) = X$.

Prove that every subspace of a metric space with Property P also has Property P. [2]

THE END